

Evacuation Analysis in Deep Underground Multi-Level Buildings Using an Agent-Based Model

Jongwon Lee

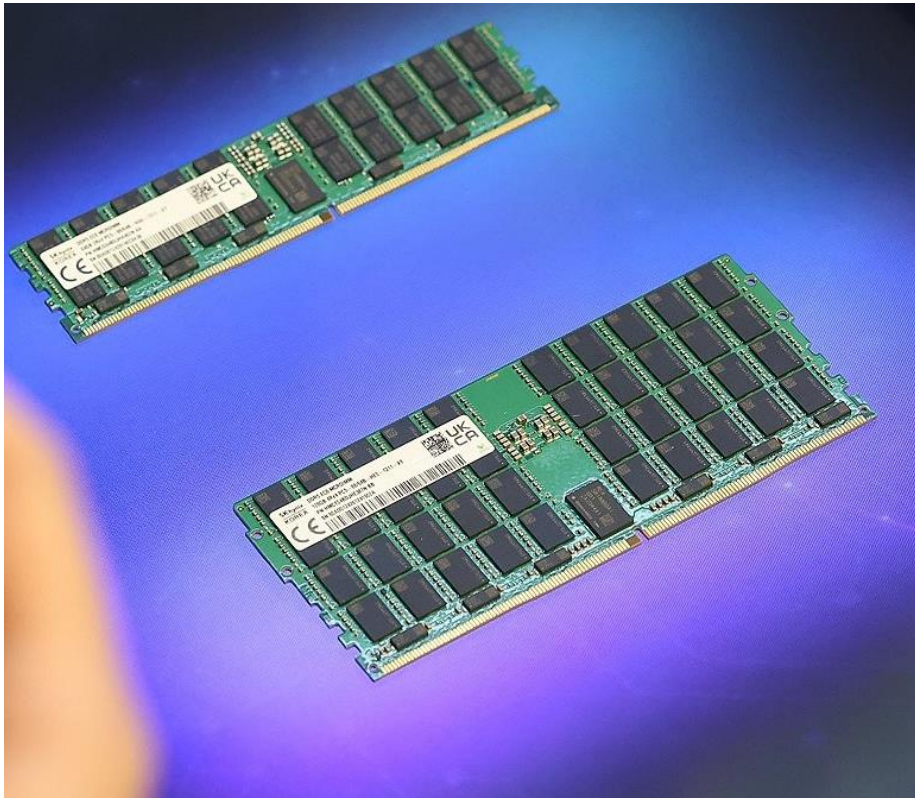
Self-Introduction



I love sports and working out.
Hiking, snow sports,
weight training...
And I love animals.
They are my cats.

Ph.D. student

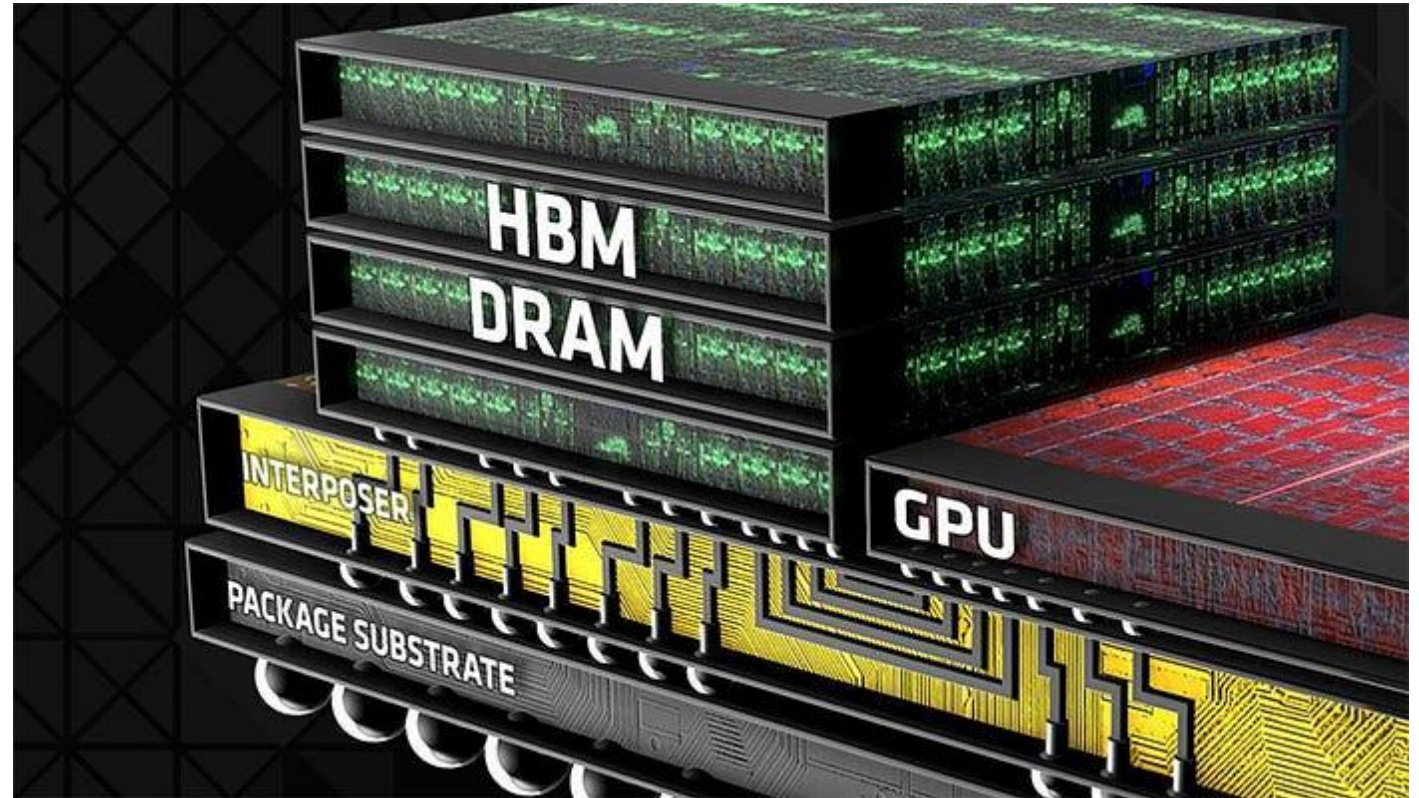
Intro of Introduction



Source:YonhapNEWS

DRAM; Dynamic Random Access Memory

2D Memory
Horizontal Scaling



Source:medium.com/@junyoungshin0122

HBM; High Bandwidth Memory

3D Stacked Memory
Vertical Stacking

Intro of Introduction



Source: startour

Los Angeles

Sprawl City
Car Oriented Develop



Source: getyourguide

Manhattan

Compact City
Transit Oriented Develop

Background and Needs



Source:PDjournal

Fire on Subway In Daegu

- 2003. 2. 18.
- Rapid smoke propagation following the arson attack
- Fire spread to the train on the opposite track
- Evacuation failure and insufficient guidance after the train doors closed



Source:courtley

London Kings Cross Fire

- 1987.11.18.
- Fire spread through wooden escalators
- Heat trapped within the passageways, accelerating fire spread
- Evacuation delays caused by blocked escape routes



Source:Seoul

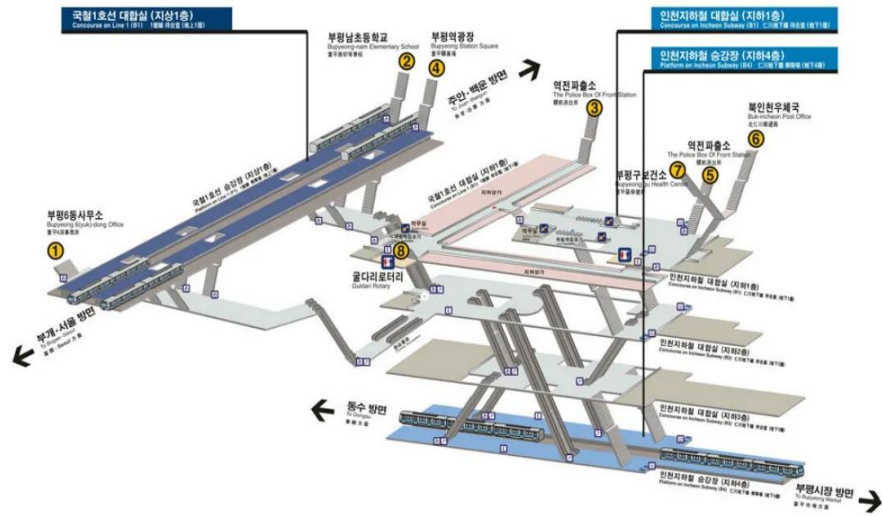


Source:GTX-A

Underground disasters are more hazardous

- Confined spaces
- Narrow and elongated passageways
- Limited vertical evacuation routes

Background and Needs



Source:Incheon Transit Corporation



Source:Yonhap News Agency



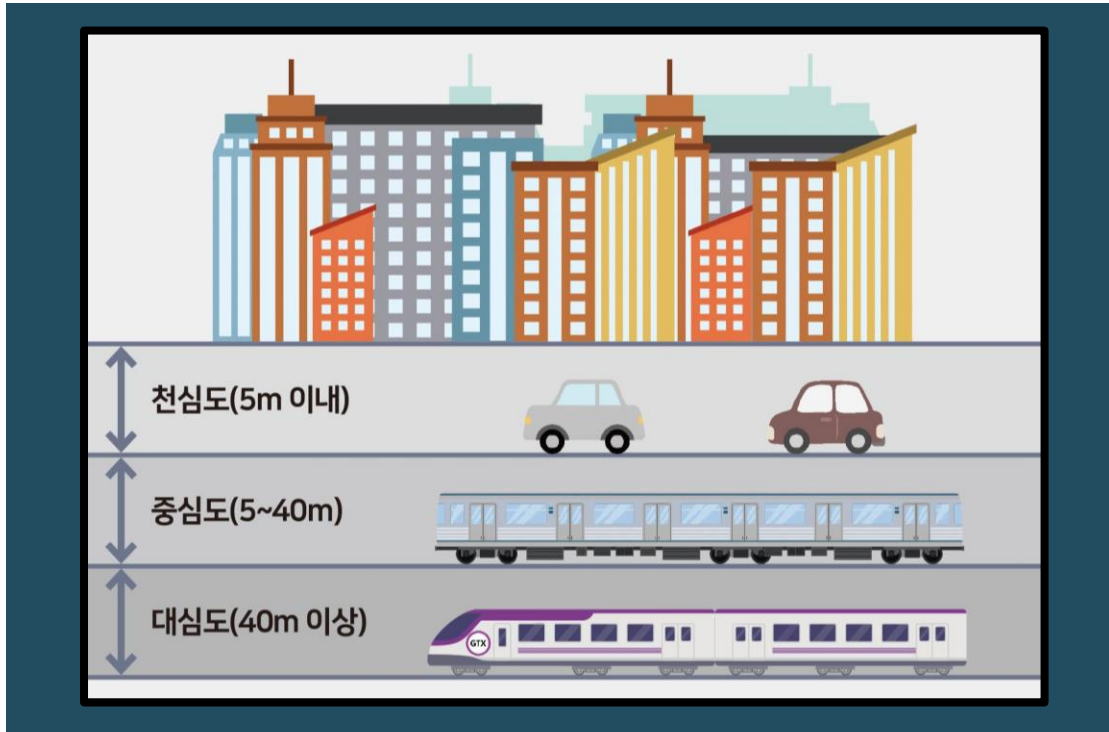
Source:advancedct

Complex building layouts

Severe crowd congestion

Golden Time

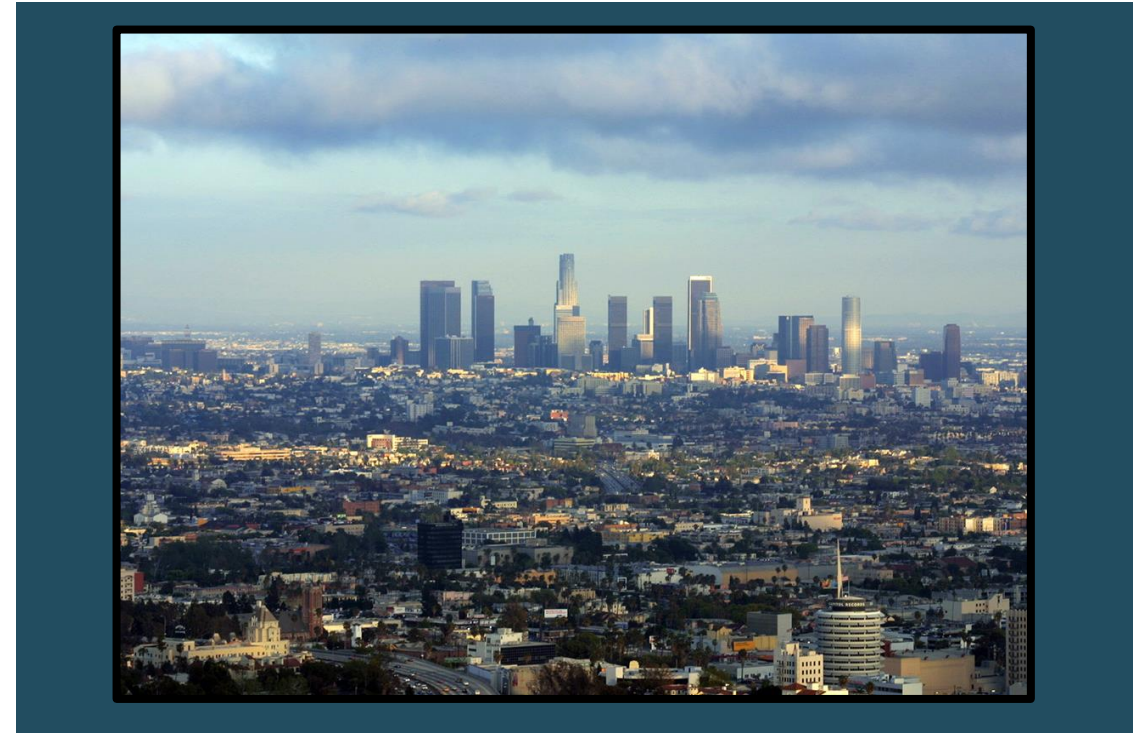
Background and Needs



Source: Hansung Univ. News

Deep underground development

High-density urban development;
High-rise, deep underground, and TOD developments



Source: Thomas Pintaric

Urbanization and population concentration

Large-scale evacuation during disasters;
Bottlenecks and panic-induced crowd collisions

The Korean Context



Source:Seoul

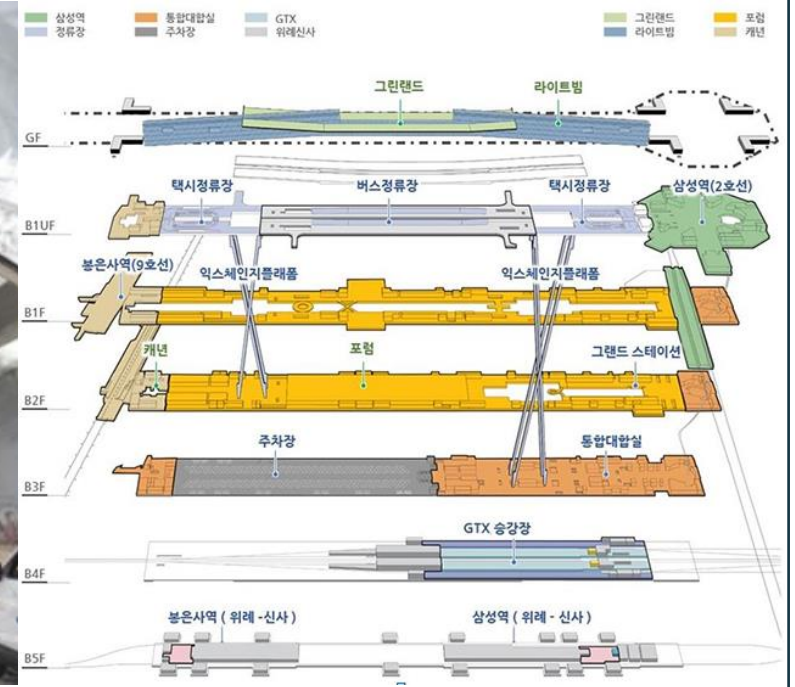
GTX



Source:Busan

Tunnel in Busan(9.6km, 5.96mile)
average 65m(213ft), deepest 120m(393ft)

Developing deep underground in transportation



Source:Seoul

Complex transfer center

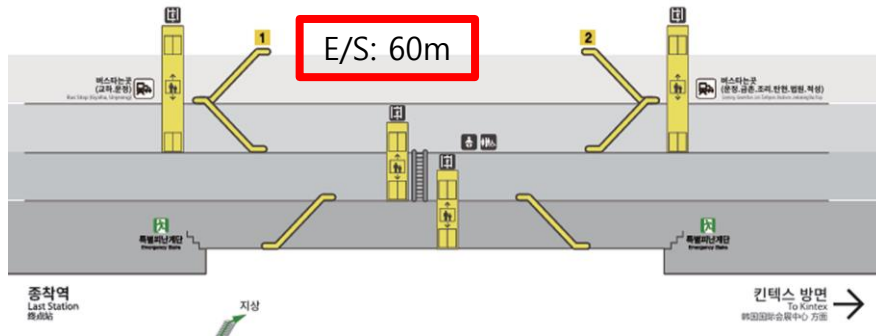
Challenges in Deep Underground Buildings



Instagram reel

Running crew every morning

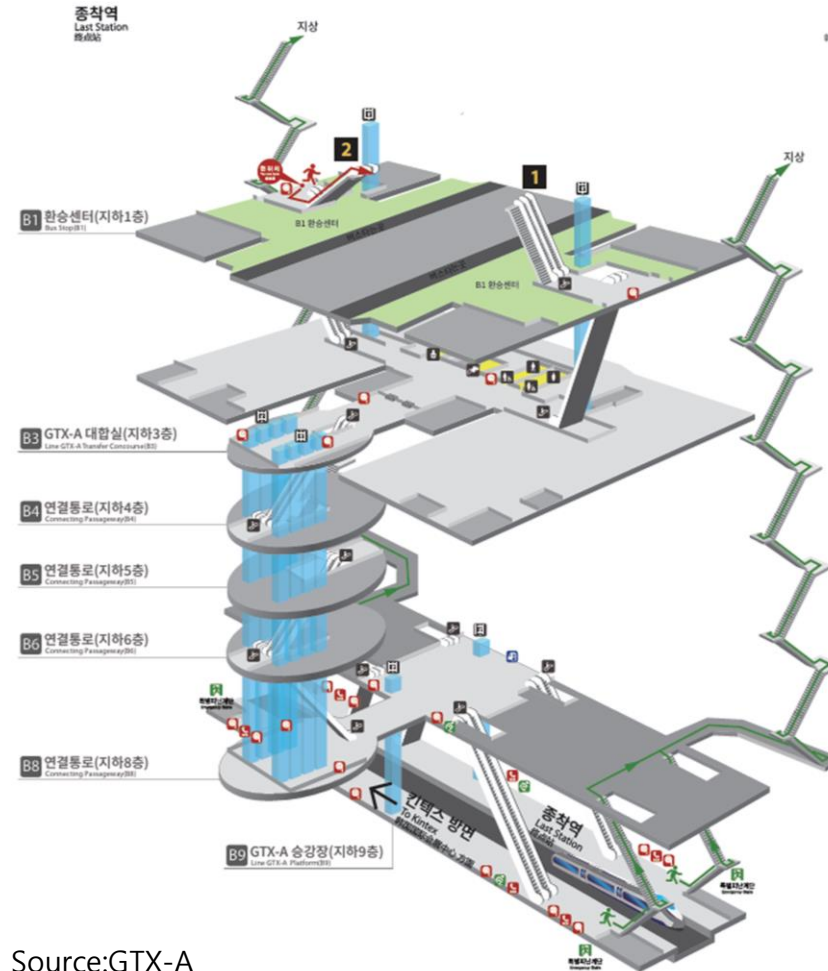
Challenges in Deep Underground Buildings



Source:GTX-A



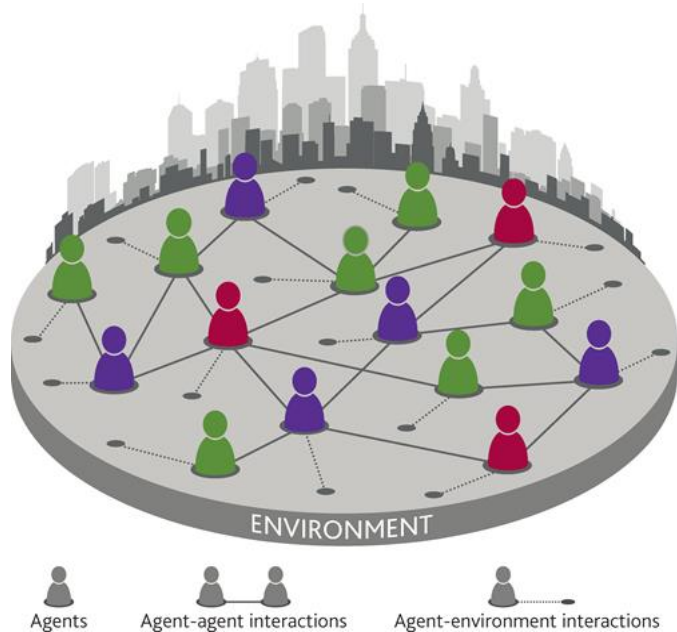
Source:ohmynews



Source:GTX-A

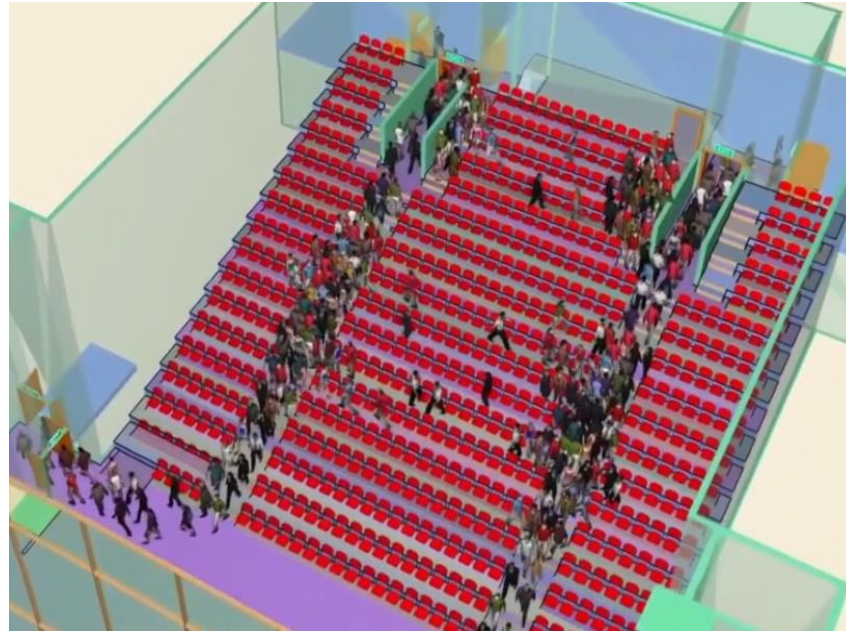
- Long vertical travel distances
- Dependence on elevators
- Bottlenecks and loss of directional awareness

Topics of Interest



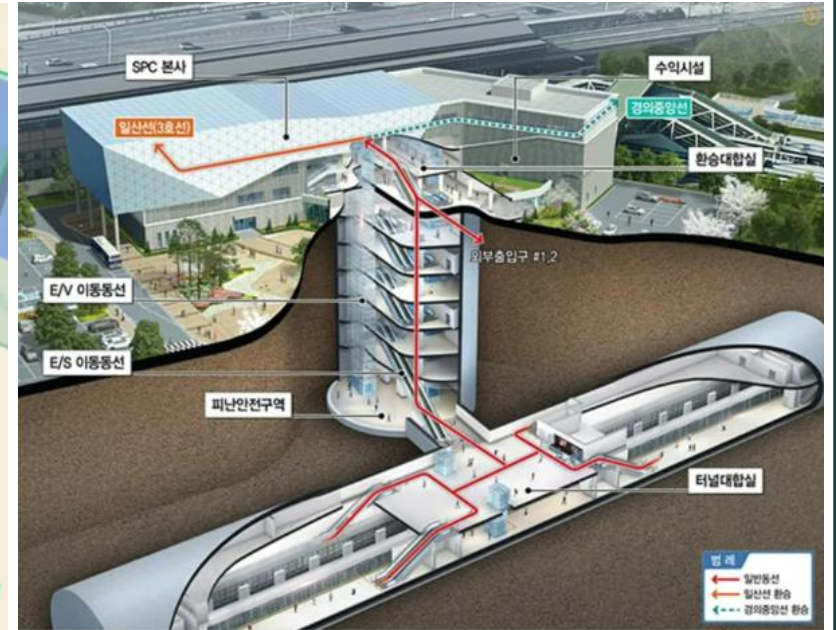
Source: Bank of England

Agent-Based Model



Source: SHEVS IFT Consultants

Evacuation simulation



Source: Seoul

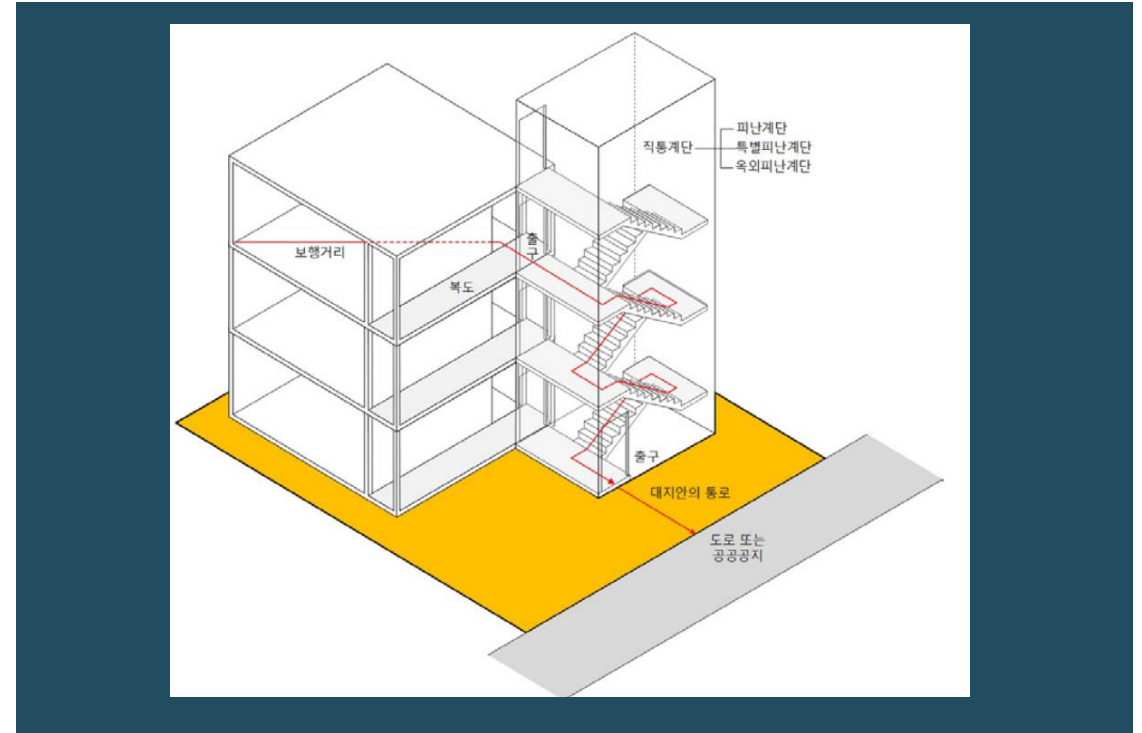
Deep underground station

Need for Simulation



Source: Milad H. & Majid S. (2017)

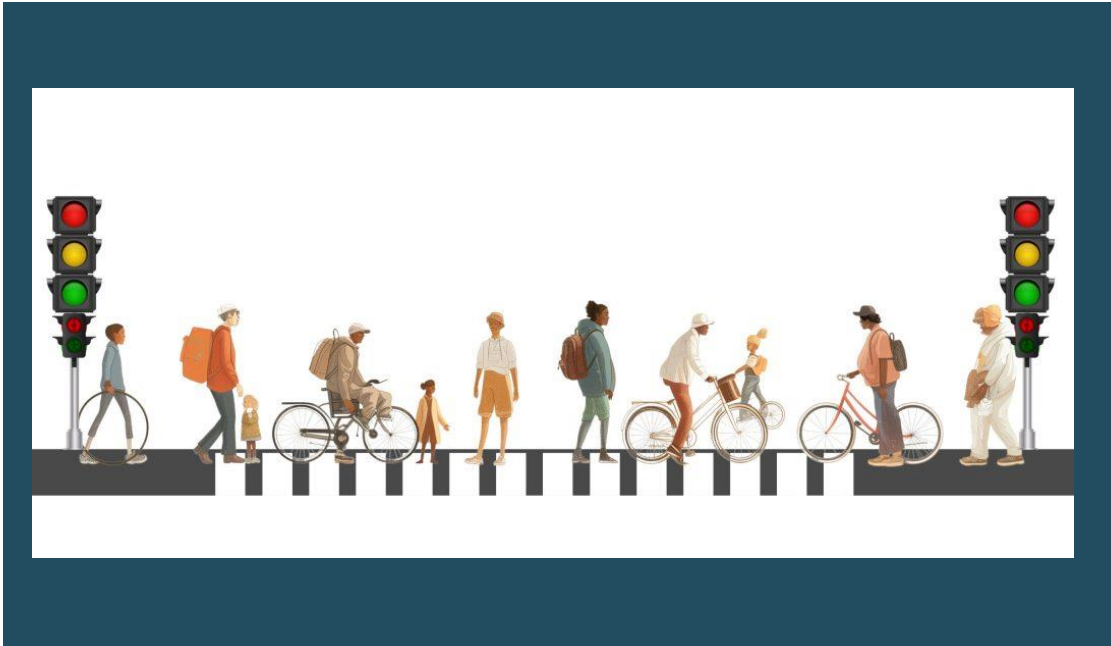
dangerous, cost, repeatable



Source: midasCAD

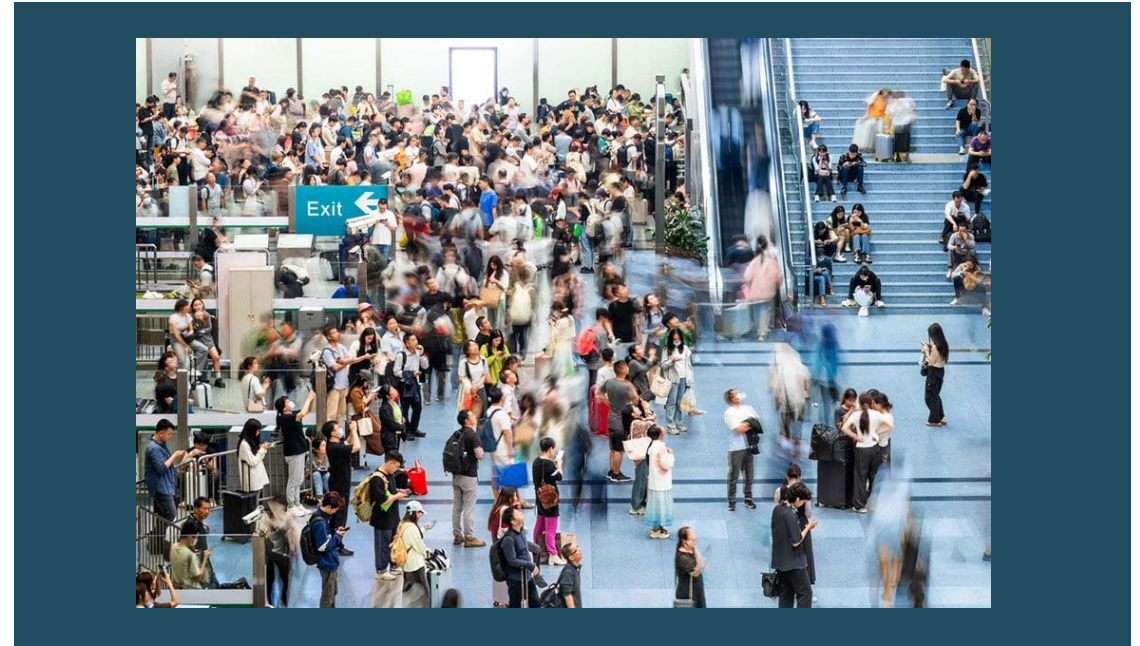
Evaluate the effects
by modifying physical building elements

Why ABM Is Needed



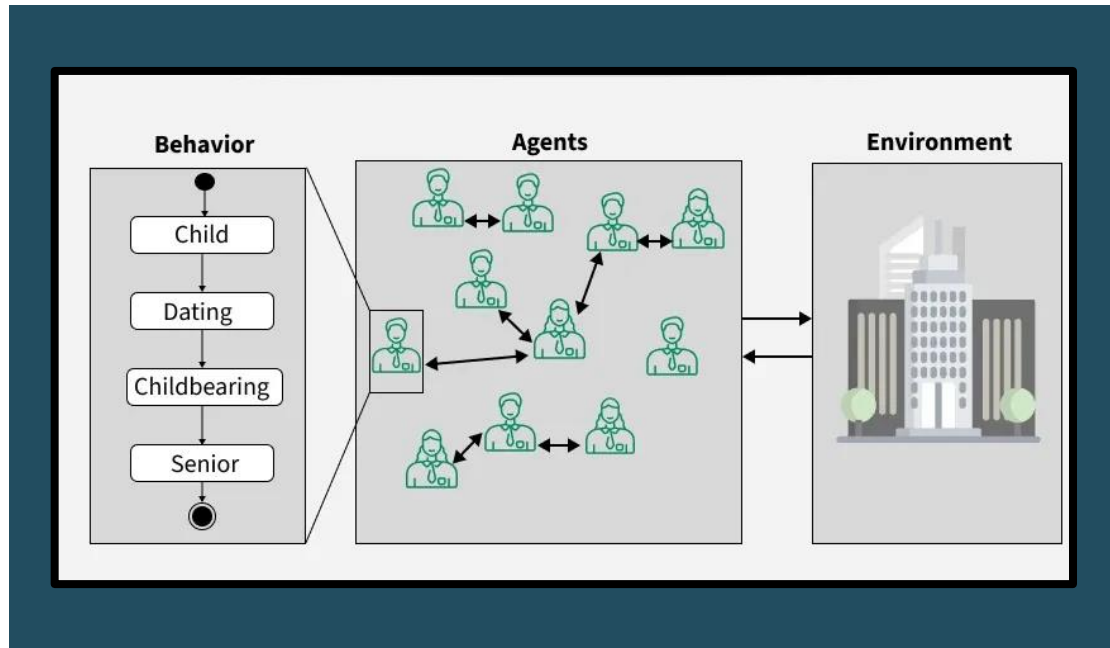
Source: passnownow

Individual behave differently



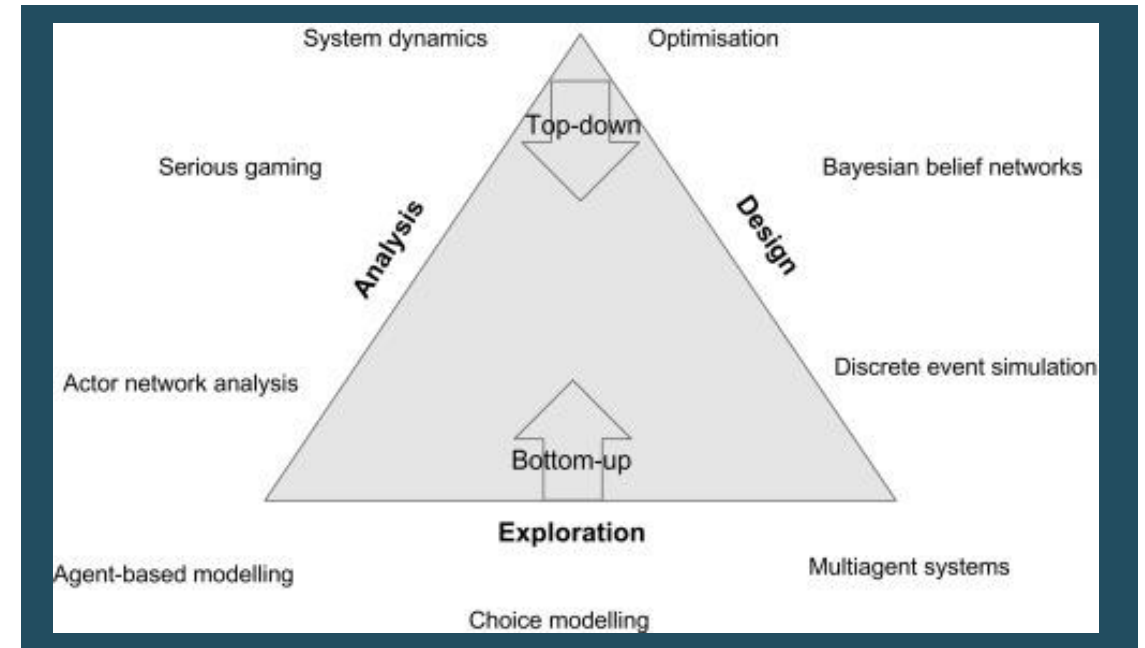
Walking speed, waiting behavior,
exit selection, route choice

Definition of ABM



Source: passnownow

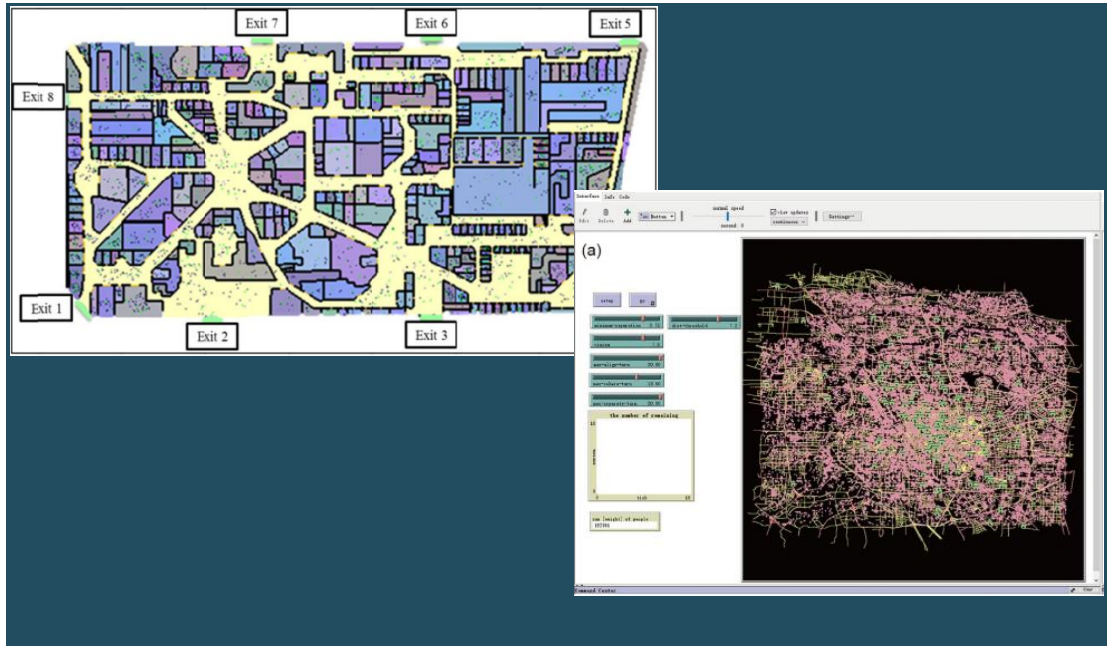
Model individual behaviors,
Analyze emergent system behavior



Source: Émile J.L. Chappin et al. (2020)

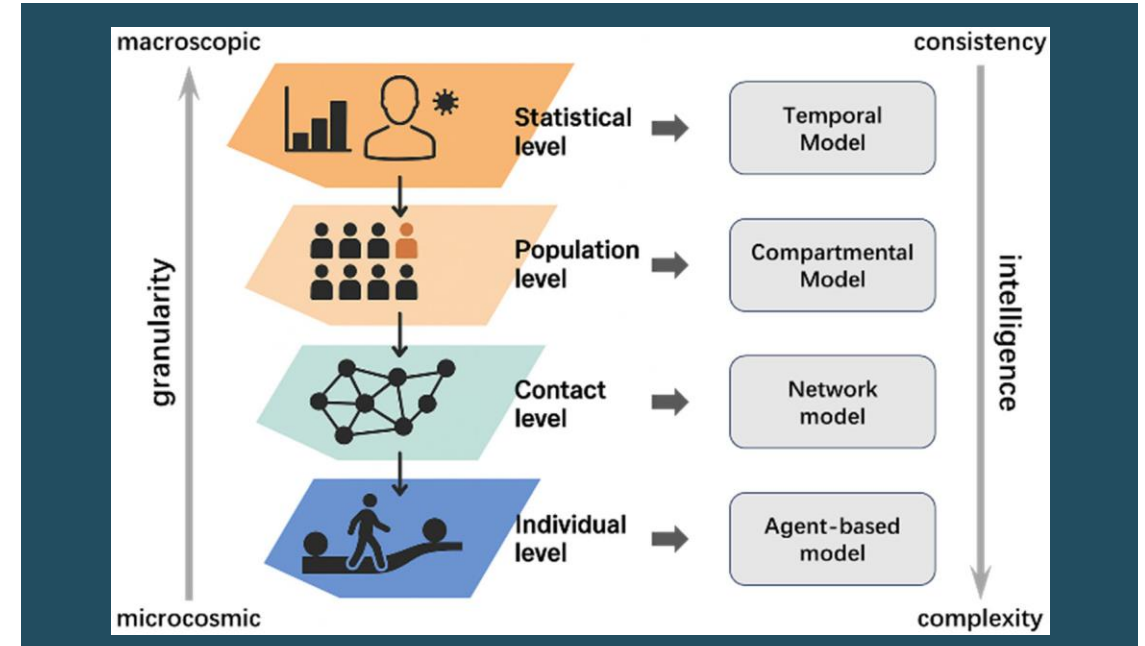
Bottom-up

Applications in Urban Studies



Source:Chiwon Ahn et al.(2016), Yang et al.(2023)

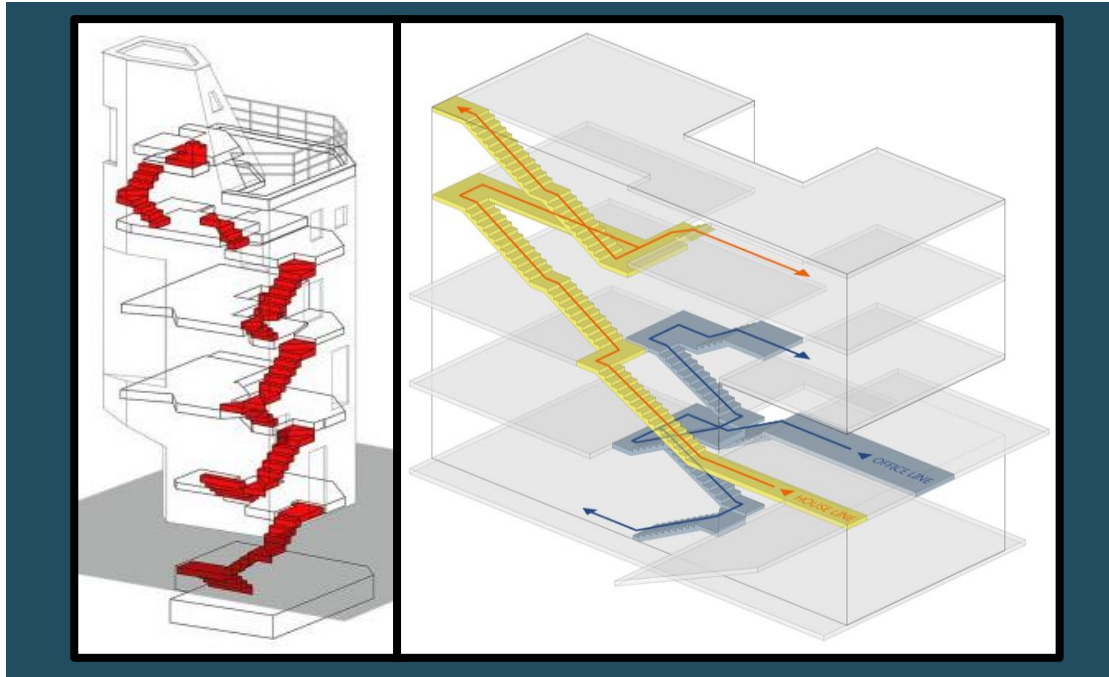
Transportation, pedestrian movement, disasters, infectious diseases, urban planning, and commercial districts



Source:Xiangyu Zhang

- Urban problems are driven by individual behaviors and interactions.
- The more complex the phenomenon, the greater the value of ABM

Specific Research Topics



Source:brique

3D evacuation in vertical spaces

- Characteristics of inter-floor evacuation
- Bottlenecks, elevators and escalators,
- mobility-impaired individuals

서울 지하철 비상대피시간 초과역사(상위 3개역사) 및 대피시간						
구분	초과 시간					
	1순위 ¹⁾	대피시간(분)	2순위	대피시간(분)	3순위	대피시간(분)
1호선	-	-	-	-	-	-
2호선	이대	8.24	을지로 4가	6.41	영등포구청	6.02
3호선	옥수	8.91	고속터미널	7.21	가락시장	6.59
4호선	남태령	7.62	쌍문	6.18	-	-
5호선	영등포시장	11.84	신정	9.68	여의나루	8.19
6호선	버티고개	9.24	창신	7.48	녹사평	7.17
7호선	송실대입구	10.6	반포	9.44	중화	8.81
8호선	산성	9.96	-	-	-	-
9호선	-	-	-	-	-	-

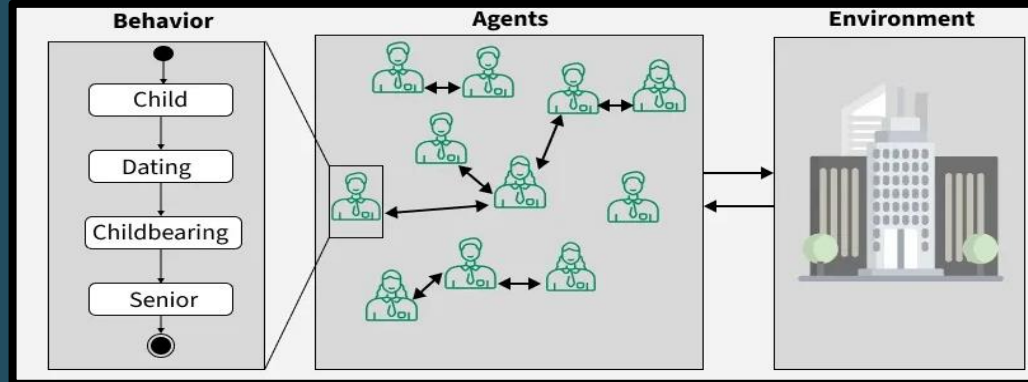
Source:CookieNews

Evacuation performance under different scenarios

- Availability of vertical transportation systems (Elevators/Escalators)
- Disaster occurrence timing, performance indicators, Congestion

Related Theories

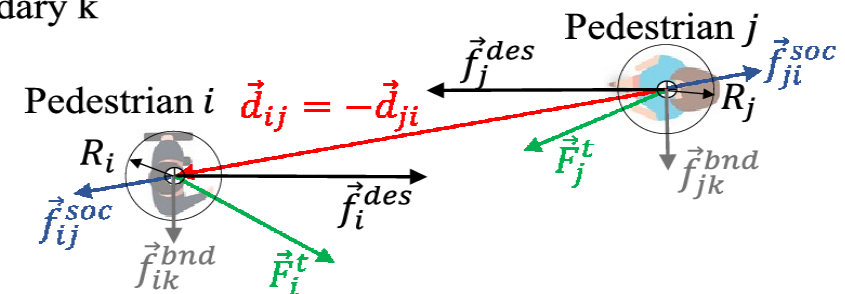
Agent-Based Model



Source: geeksforgeeks

Social Force Model

Boundary k



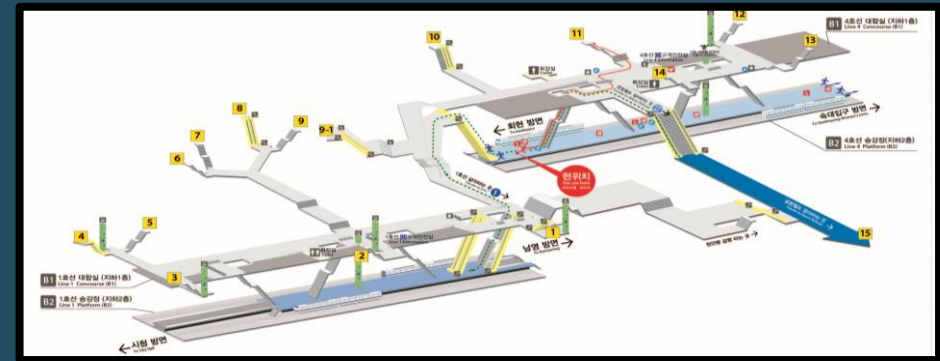
Source: Jafari, A., & Liu, Y. C. (2024). Time-to-Collision Based Social Force Model for Intelligent Agents on Shared Public Spaces. International Journal of Social Robotics, 16(9), 1953-1968.

Space modeling



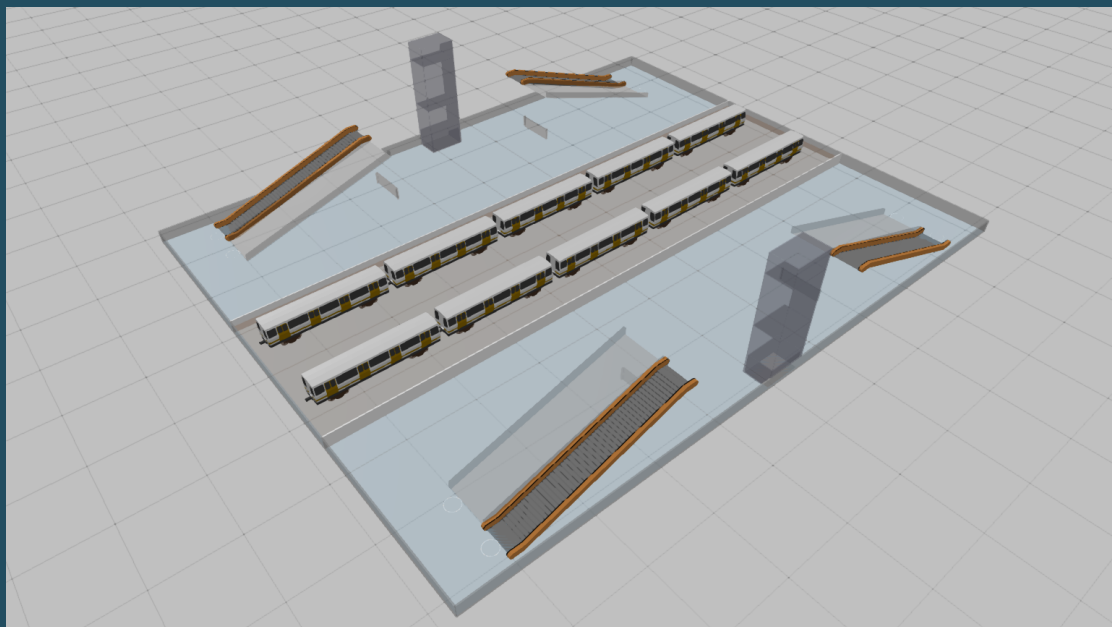
Source: Seoul Metro

Network construction

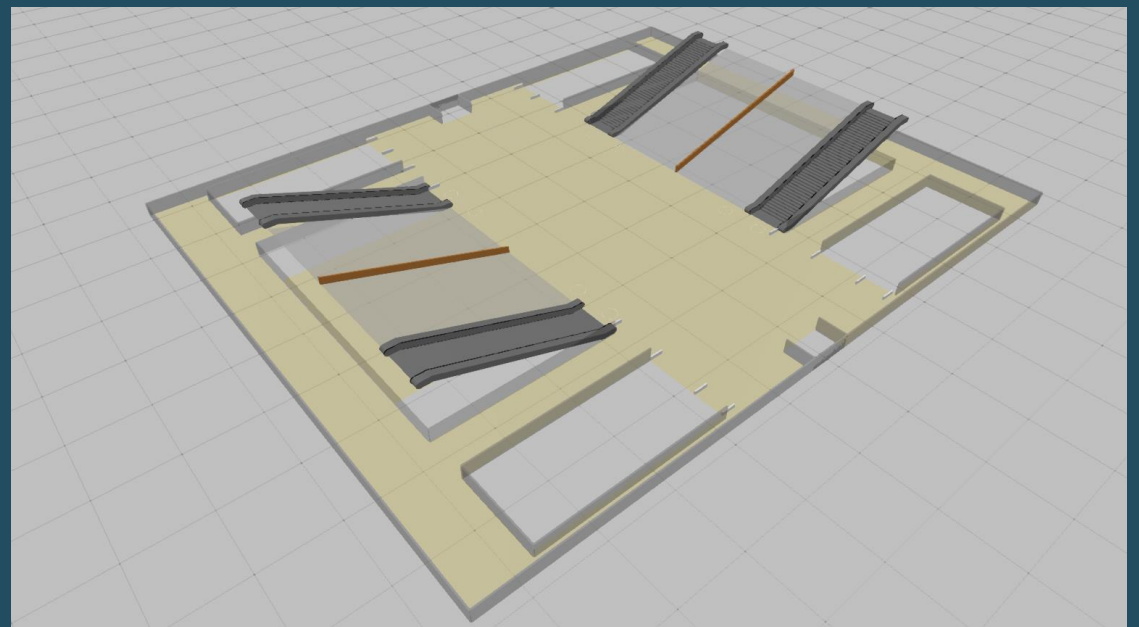


Source: Seoul Metro

Research Methodology

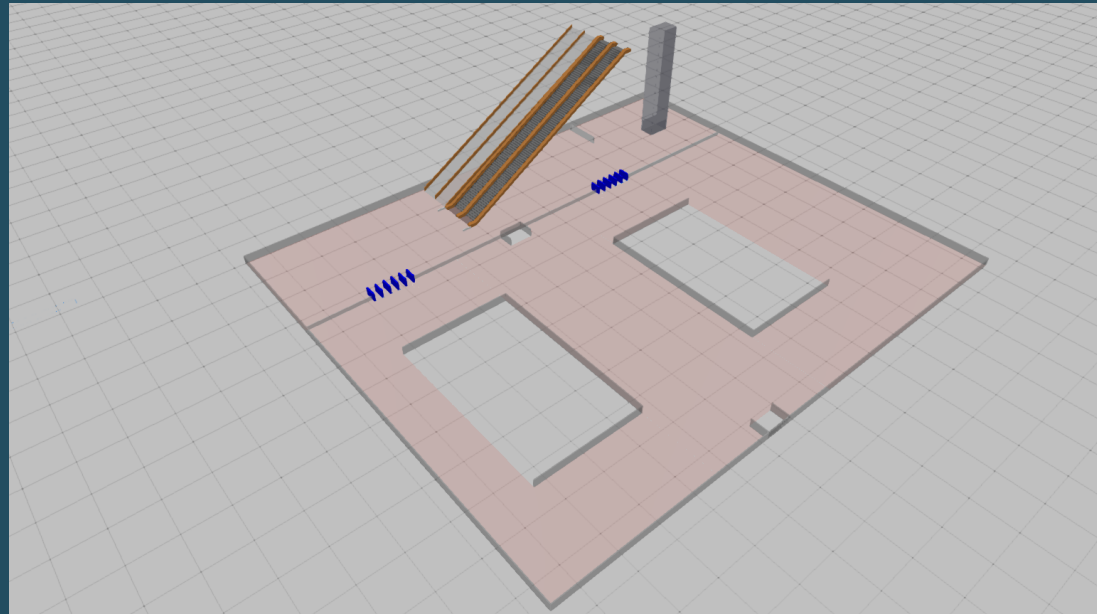


B3
E/S, stair, E/V

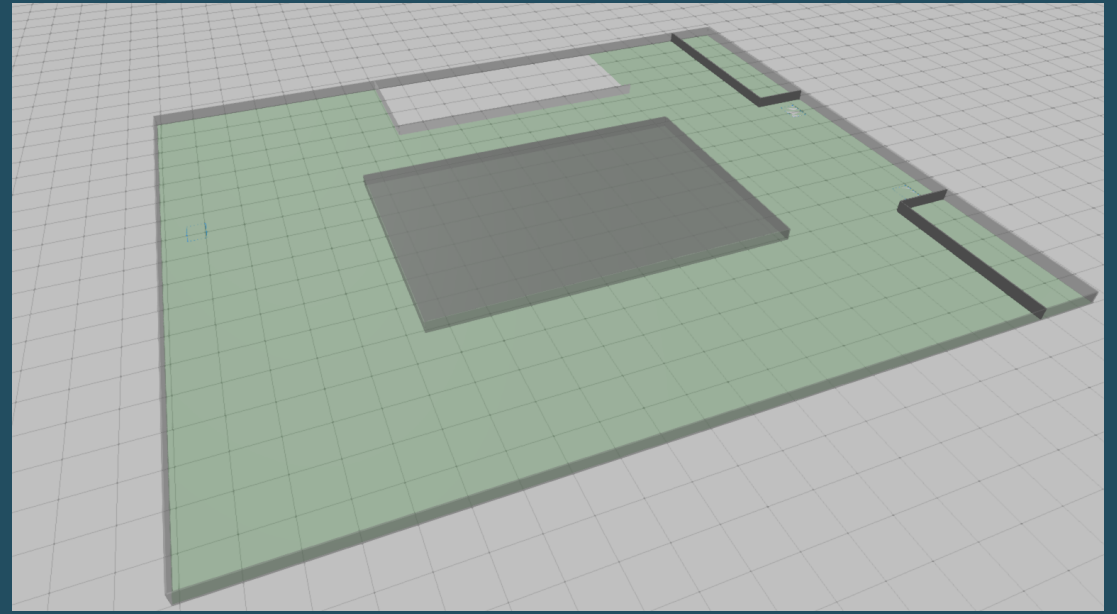


B2
E/S, stair, E/V

Research Methodology

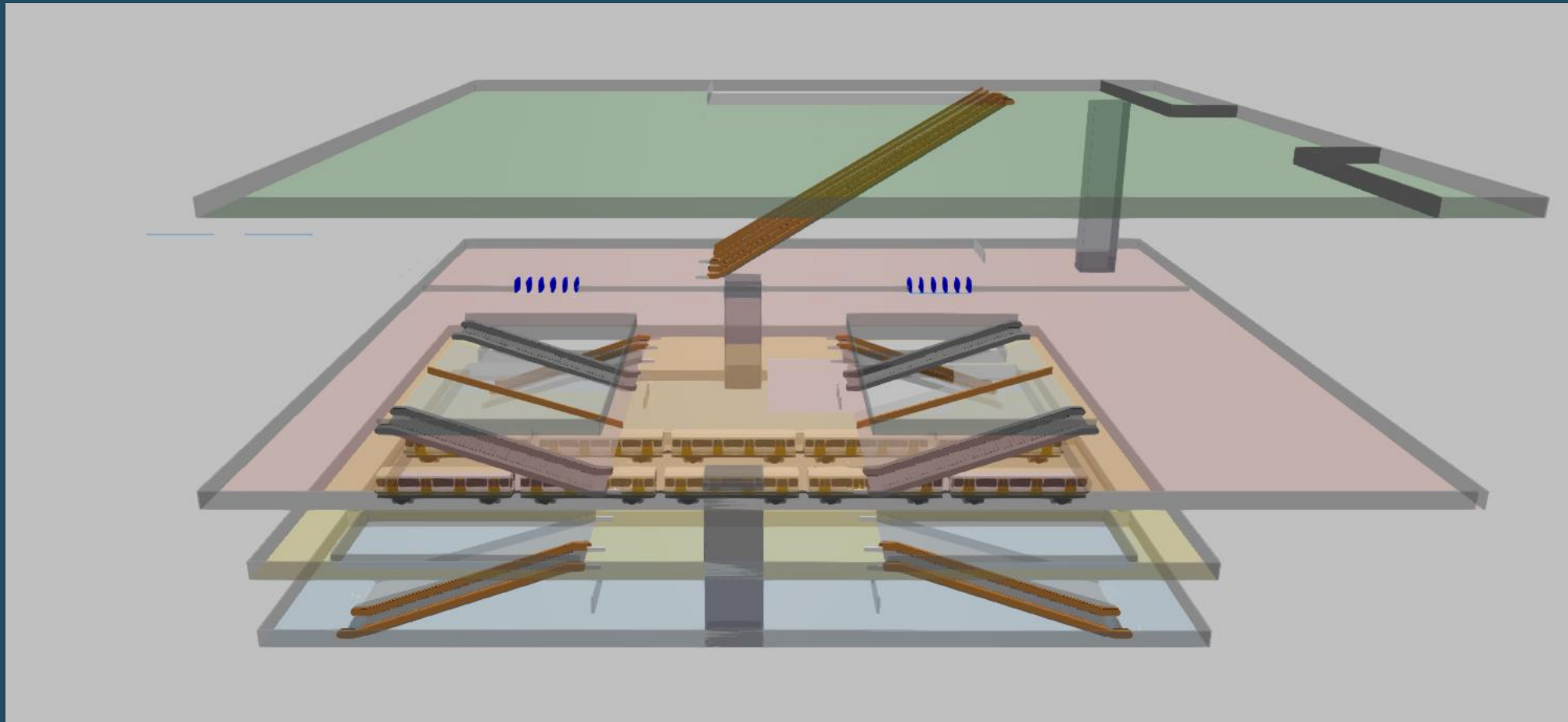


B1, subway gate
Subway gate, E/S, stair, E/V

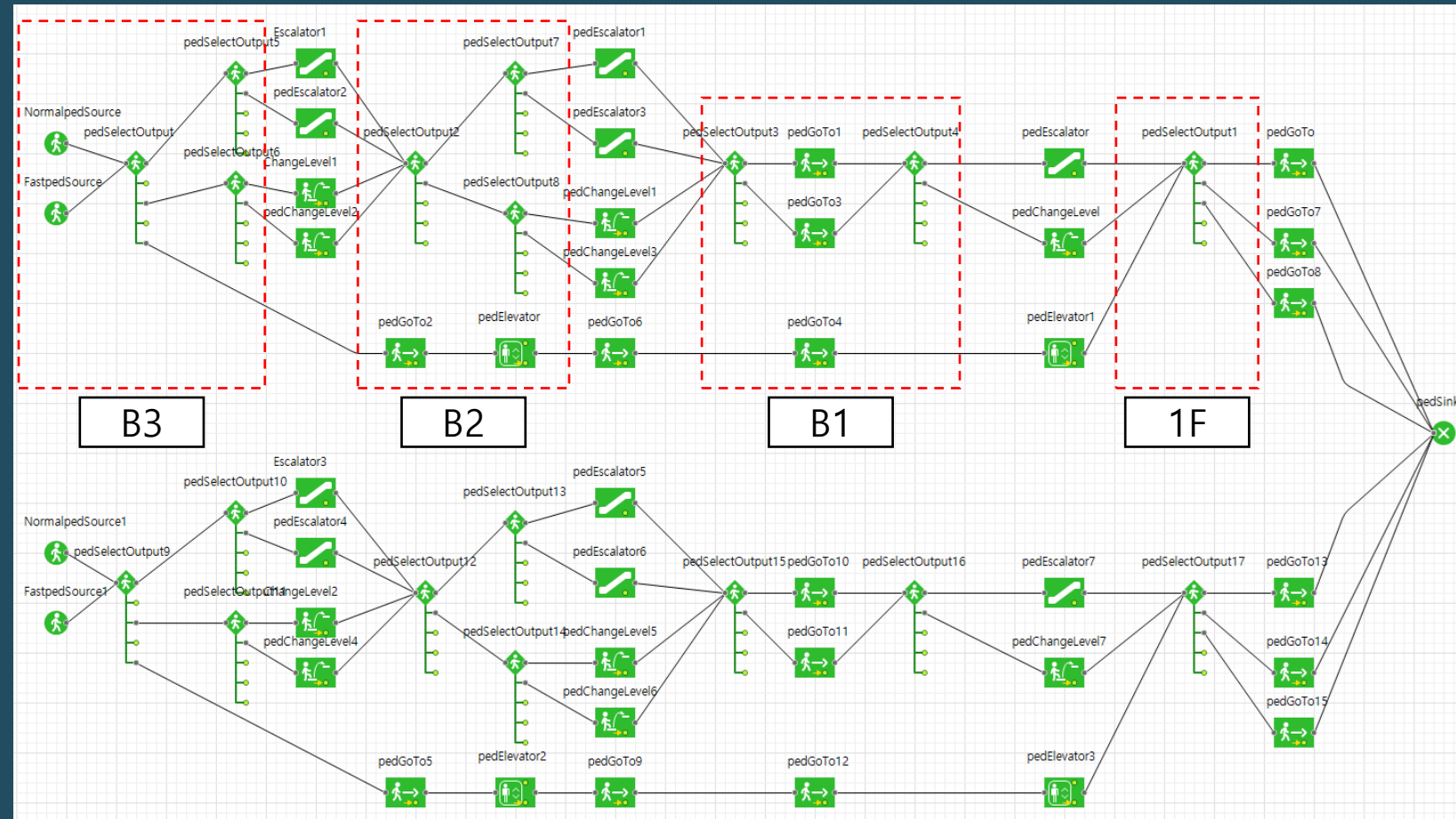


1F, exit
E/S, exit

Research Methodology



Research Methodology

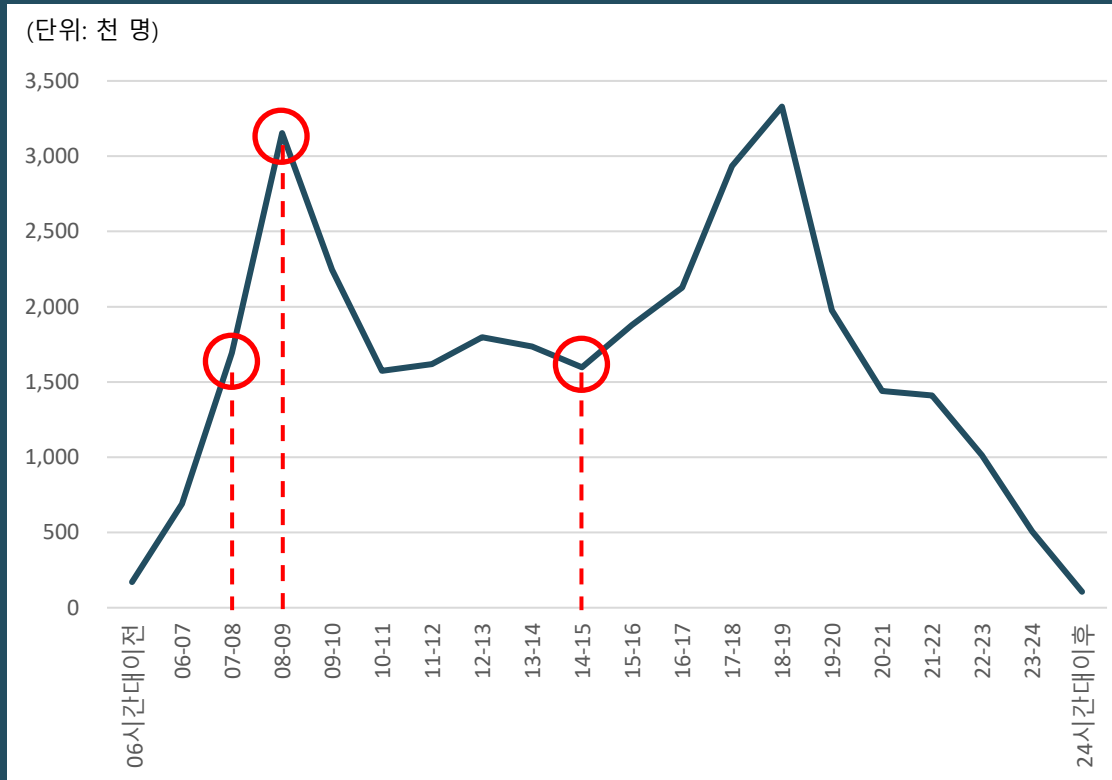


Flowchart

Research Methodology

Using Seoul station data

Passenger of Line 1 and 4 at Seoul station



Data

Rate of E/V user
Passenger of Line 1 and 4 at Seoul station
Train arrival schedule

7-8 am

Rate of E/V user: 5% 258 person

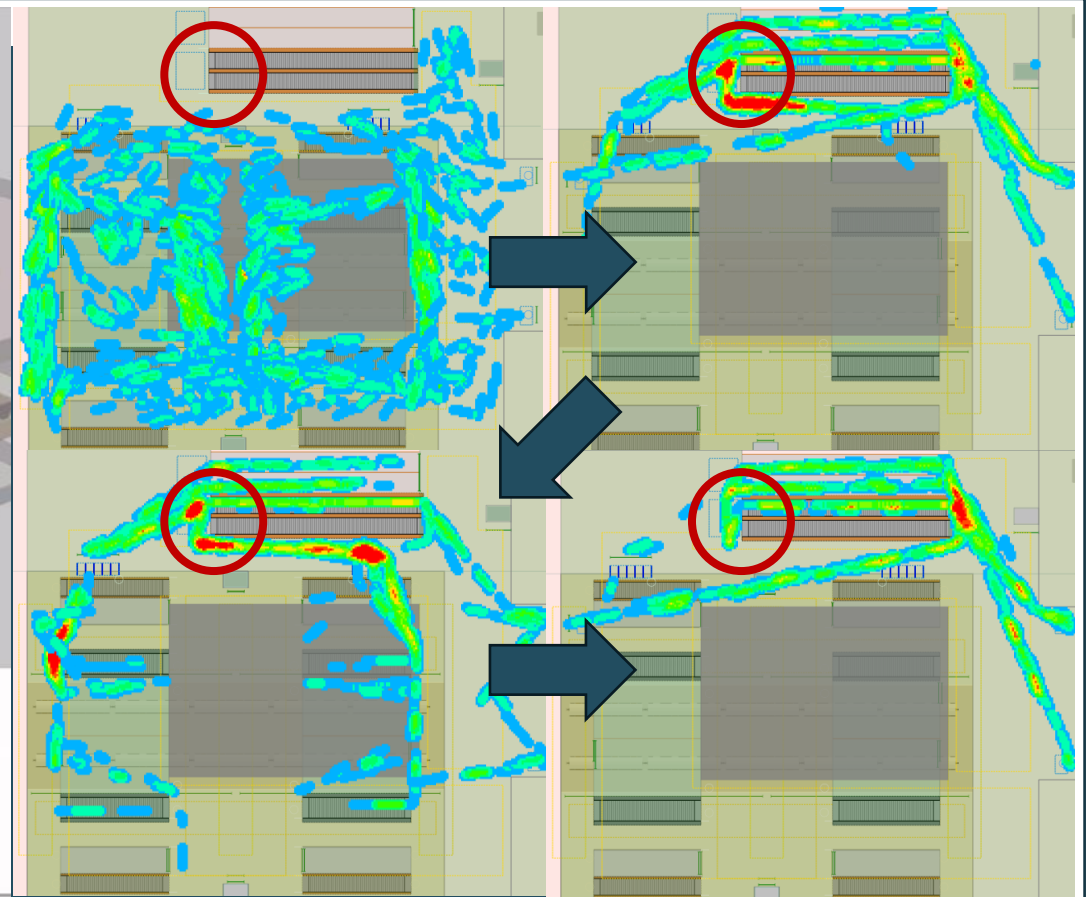
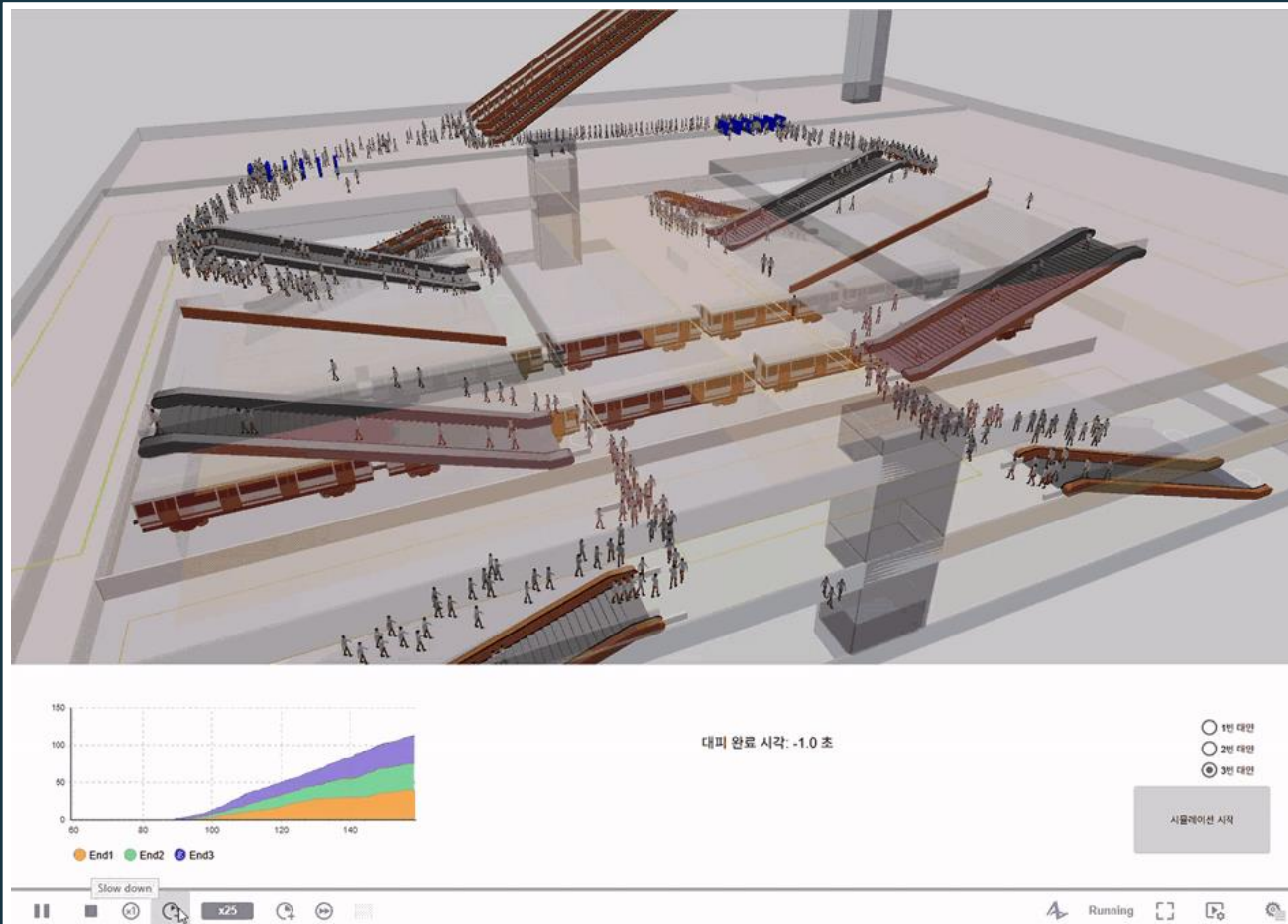
8-9 am

Rate of E/V user: 3% 319 person

2-3 pm

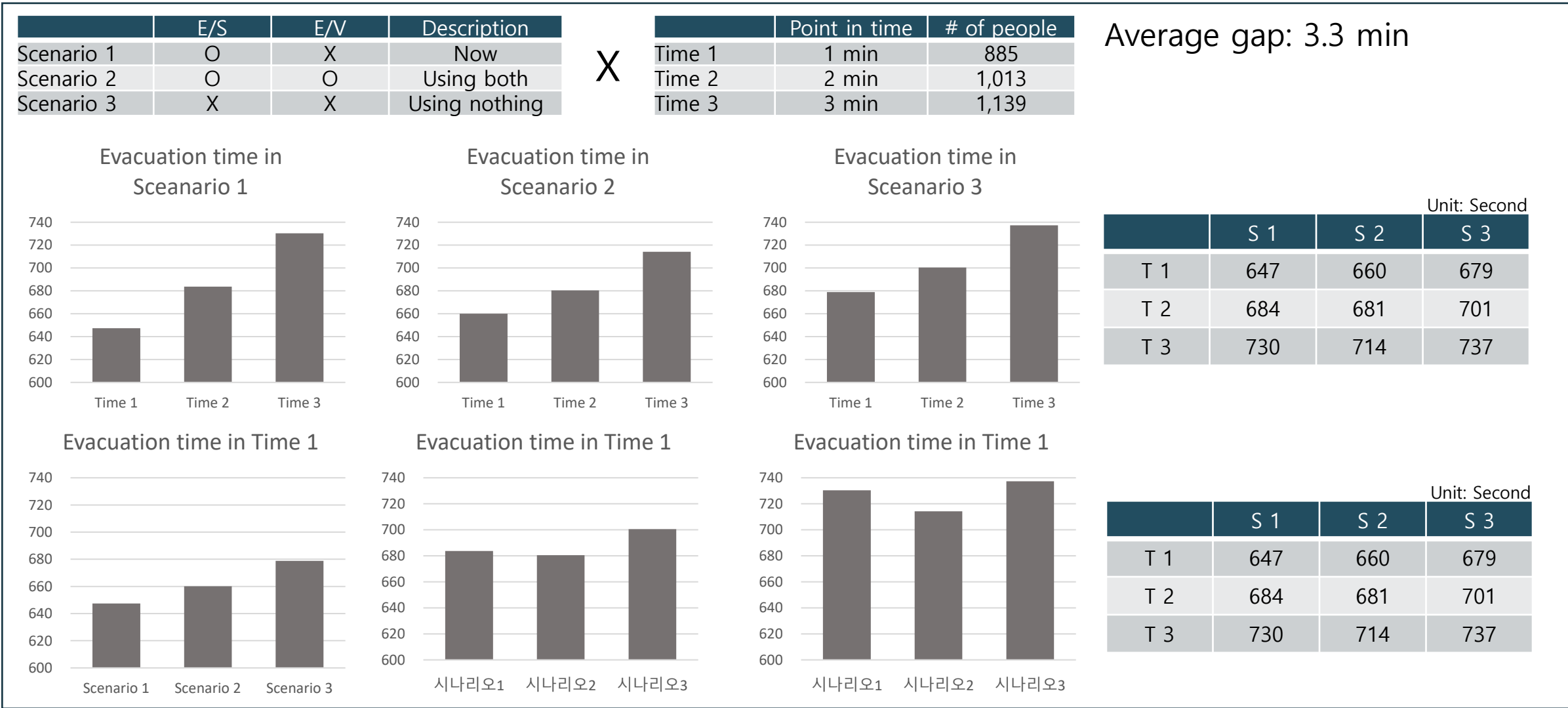
Rate of E/V user: 12% 420 person

Research Findings



The bottleneck lasted for four minutes

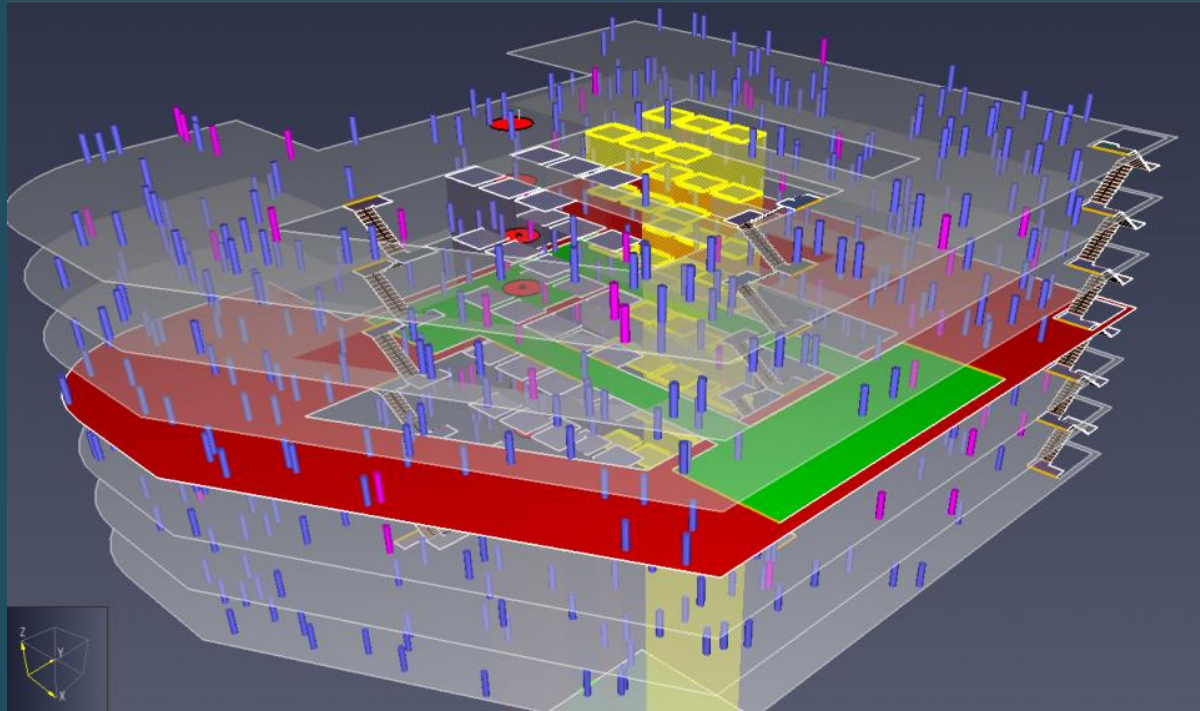
Research Findings



Unit: Second

Unit: Second

Research Extension: Evacuation Stream Control System



Source:Thunderheadeng

Effectiveness of Refuge Floors

- Emergency rescue through refuge floors in high-rise buildings
- Refuge sections improve the efficiency of multiple high-speed elevators
- Combined elevator and stair evacuation via refuge floors is the optimal evacuation strategy

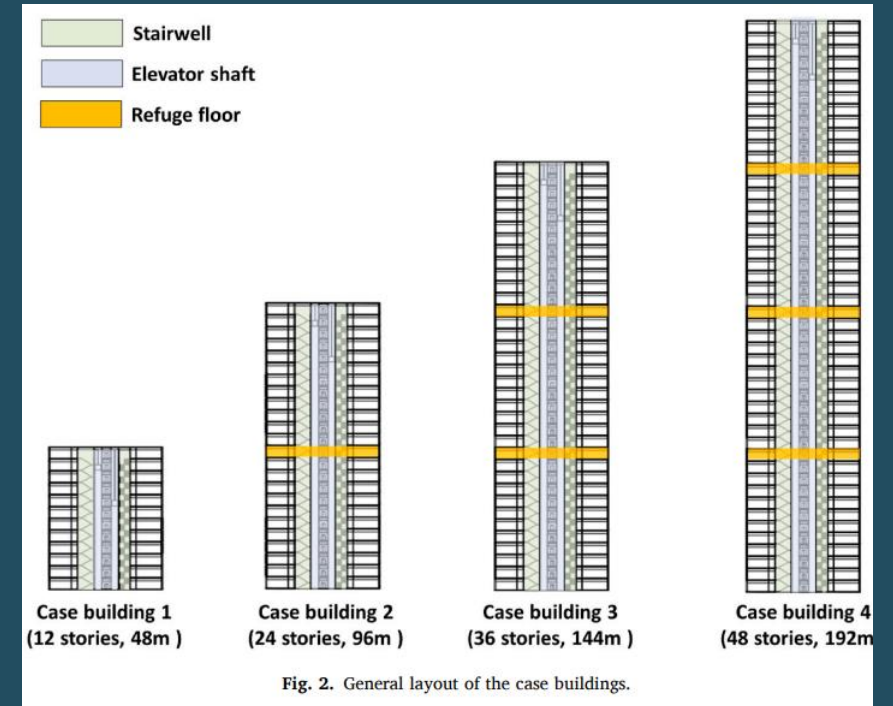


Fig. 2. General layout of the case buildings.

Source:Fang, H. et al. (2023)

Research Extension: Evacuation Stream Control System

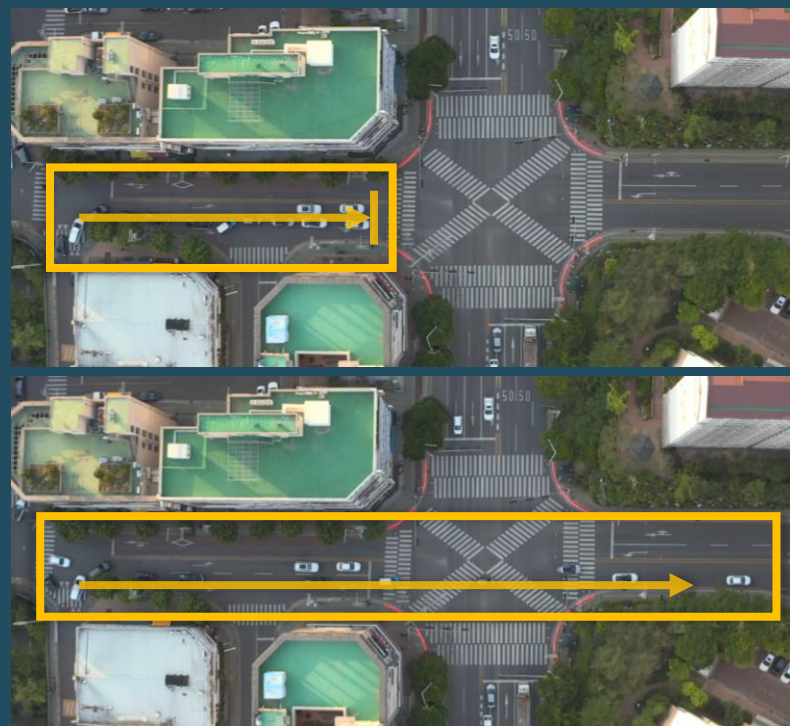


Source:youtube@공작소

GTX-A Unjeongjungang Station is located on Basement Level 8 (B8).
Reaching the ticket gate on B3 takes more than 3 minutes by escalator but only 30 seconds by high-speed elevator.

A dedicated evacuation strategy for high-speed elevators is required

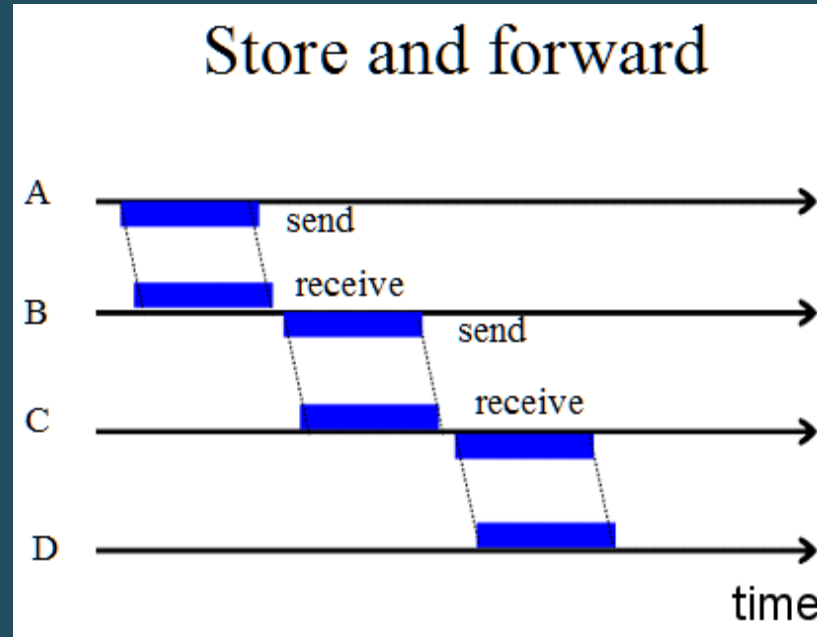
Research Extension: Evacuation Stream Control System



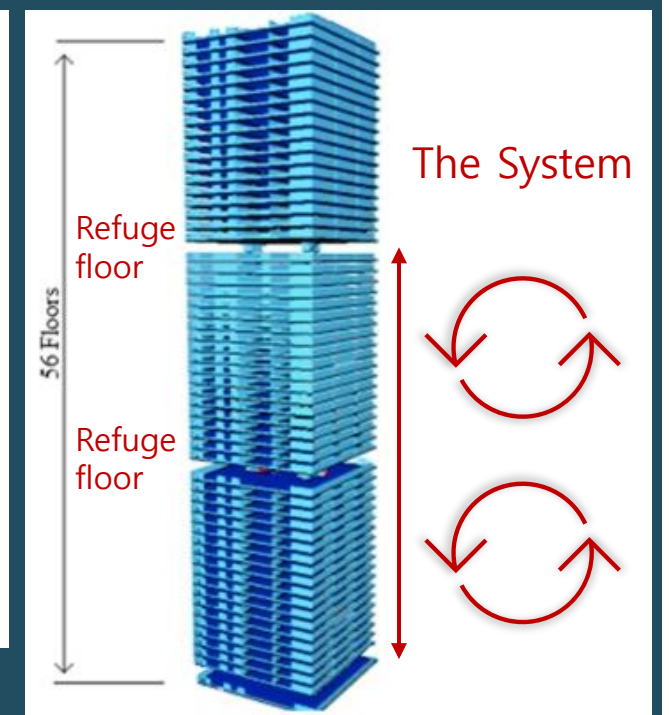
Source:youtube@쓰르륵

Store and Forward

: a communication and switching method where a network device or relay completely receives and temporarily stores data

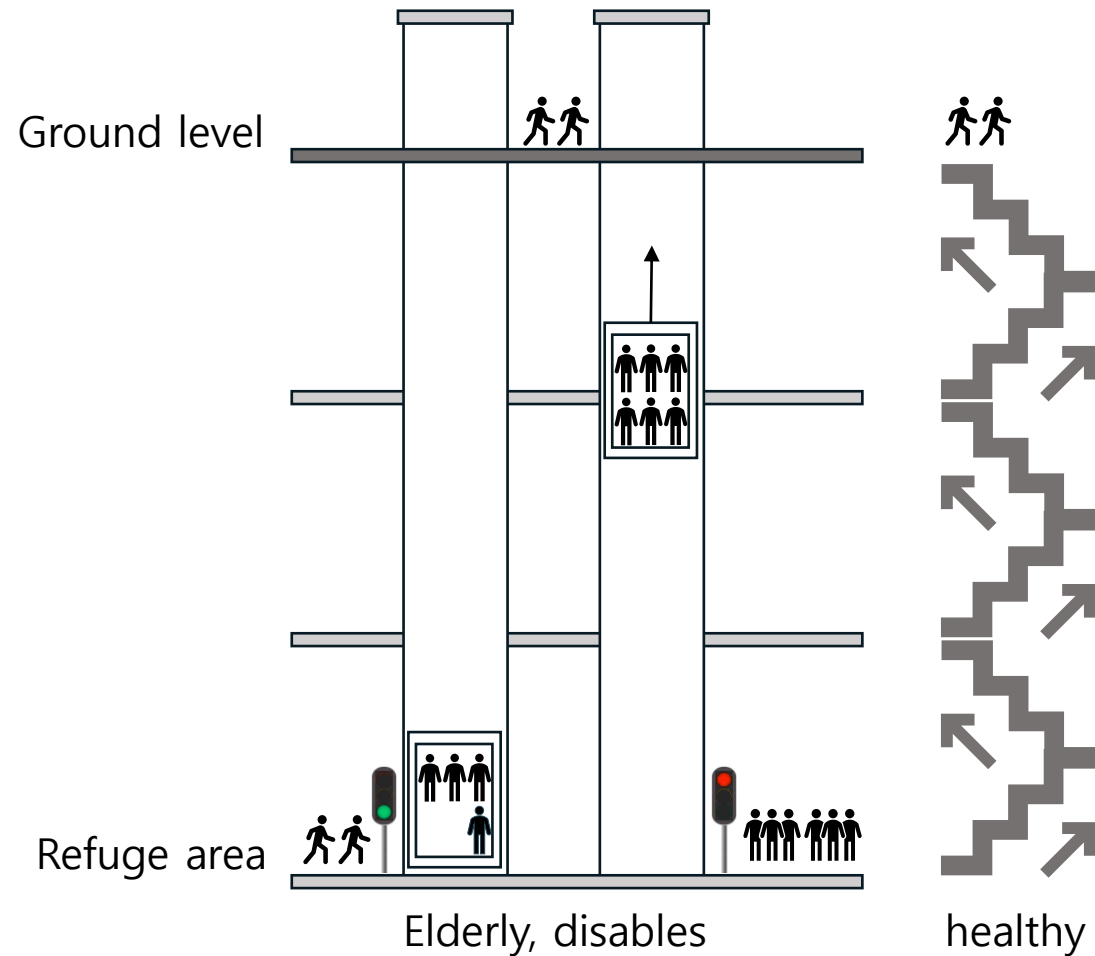


Source:remuto.tistory.com

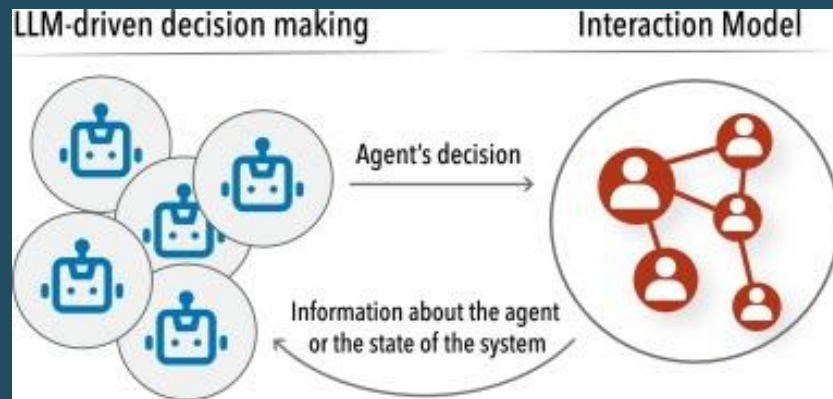


Source:Mitko Aleksandrov et al. (2019)

Research Extension: Evacuation Stream Control System



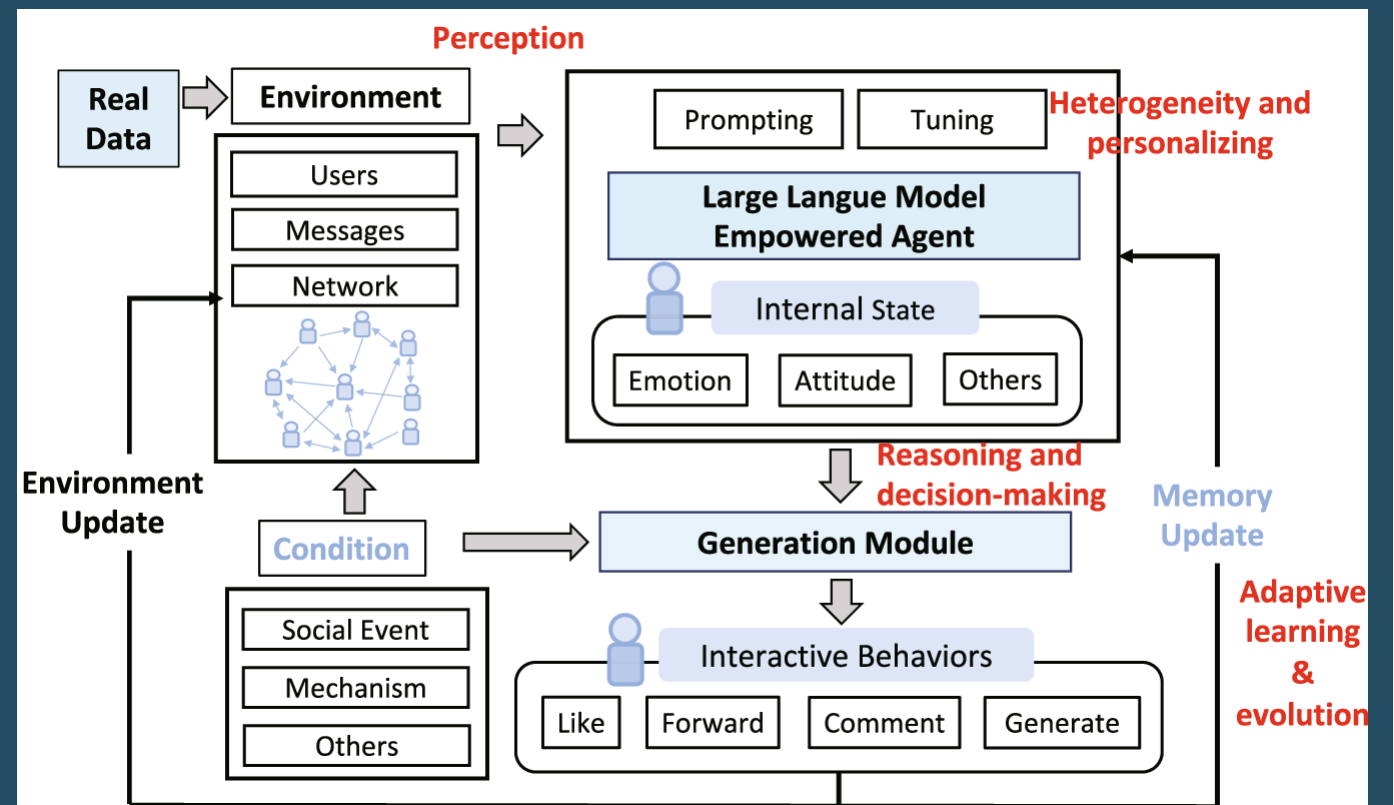
Research Extension: LLM Collaboration



Source:remuto.tistory.com



Source:aws



Source:remuto.tistory.com

Variety of agents and interactions in city

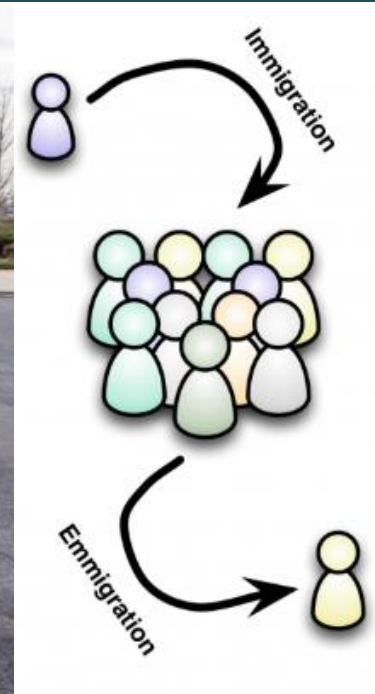
Research Extension: LLM Collaboration



Source:Ed Yourdon



Source:Amanda RP (2026)



Source:Scott F. R. & Gene B. (2018)

Variety of agents and interactions in city

Research Extension: Analysis Scope

Macro-level:

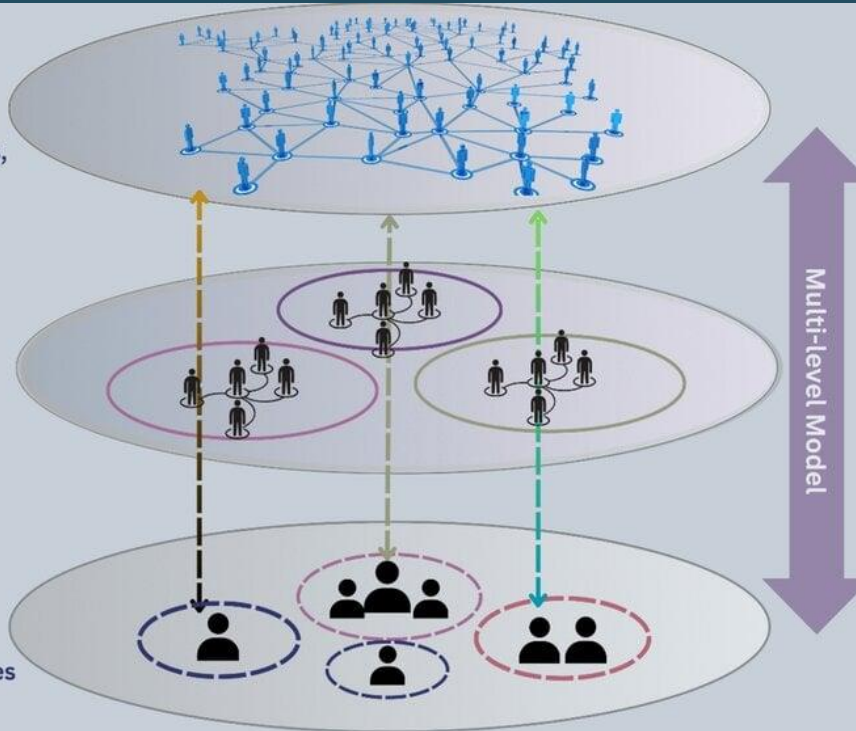
States, Legal Systems,
Economies, World

Meso-level:

Organizations,
Communities,
Ethnic Groups

Micro-level:

Individuals,
Relationships, Families



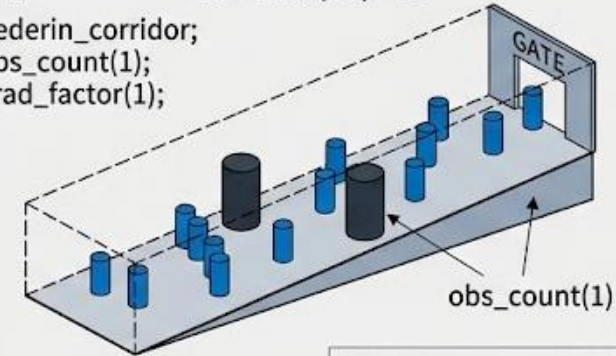
Source: mark Rayn et al. (2024)

We should focus micro analysis index
;bottleneck point, queue length, agent trajectory, timely density map etc.

ax3d_1

기존 환경 (S1) - 3D

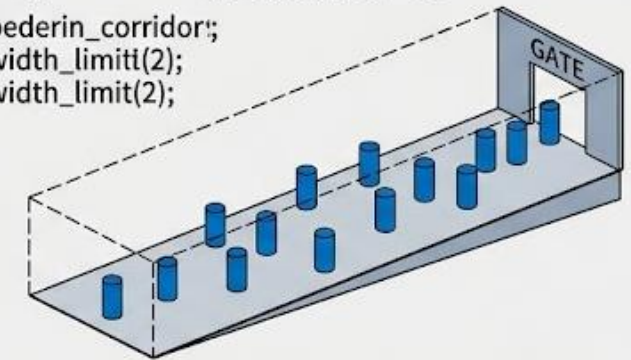
- pederin_corridor;
- obs_count(1);
- grad_factor(1);



ax3d_2

개선 환경 (S2) - 3D

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- width_limit(2);
- width_limit(2);





Thank you

Jongwon Lee